KUO, CHEN CHIH 郭晨志

M.S. in Chemical Engineering • Licensed environmental specialist

Seeking **ISEP EHS Engineer**

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Xinzhuang, New Taipei · Open to relocation / shifts
Other Exempt · Car licence · Pre-hire, from Oct 2027



Profile

NTUST Chemical Engineering M.S. candidate holding three environmental specialist licences (**wastewater, waste, indoor air**), OSH exams scheduled (OSH Management Personnel, Level B, pre-screen passed, Nov 2026; Occupational Safety Management Specialist, Level A, Mar 2027); hands-on with **hazardous materials** (lithium metal, organic-solvent electrolytes, faulty Li–$SOCl_2$ cells), lab safety and **TGA / DSC thermal-hazard assessment**, plus **dry-room** clean-area work. Seeing synchrotron equipment run for 30+ years on preventive maintenance showed me how much systems matter for safety, and I want to learn to apply that in a fab.

Experience

2025 – Present

Sustainable Electrochemical Energy Development (SEED) Center, NTUST • MOE Featured Areas Research Center

Lab Safety, Equipment & Asset Maintenance Officer

- **Rules:** wrote the TGA's PPE and ventilation rules and the TGA / DSC booking and usage rules that every user, including other labs, must follow — e.g. no DSC heating of unknowns without prior TGA decomposition data; volatile solvents only in pressure-rated pans. High-risk samples (Prussian blue, sulfides) need PI approval. Result: fewer high-risk submissions, and DSC pan bursts and contamination avoided.
- **Procurement (>NTD 2M):** sourced, accepted and logged safety-critical equipment such as glovebox O_2 sensors, plus annual service contracts and consumables.
- **Lab safety (≈60–70 people, 9 gloveboxes on two campuses):** monitor glovebox O_2 / moisture levels and handle sensor faults; run daily gas-cylinder checks and lab safety walk-throughs and brief newcomers on safety.
- **Incident response:** when a classmate cut a finger on a broken reagent bottle, I assigned roles, identified the chemical, checked the SDS and sent the student to the ER, and sealed the debris as chemical waste; afterwards I replaced expired cartridges and Hexafluorine. Lesson: incidents must be escalated — I only told a senior PhD student.
- **Hazardous materials:** handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li–$SOCl_2$ cells.
- **TGA testing service:** Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST — over 600 samples in total, including my own lab's.

Summer 2022

New Taipei City Dept. of Urban Development

Summer Intern, Housing Development Section

- Rental-subsidy eligibility checks: cross-matched property, income and insurance records from the tax bureau, land registry and NHI against legal thresholds; handled frontline public inquiries and disputes.

2021 – 2024

Tunghai University Student Housing / GEPT

Resident Assistant (2 yrs) / Exam Staff (part-time)

- Safety and dispute mediation in a 100-resident dorm; exam-paper check-in with zero errors.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker **Project lead**

Database & SOP
Established the test database and standardized high-throughput screening procedures (SOP).

Training & tech transfer
Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Communications Journal submission — battery thermal stability study
Status Under review **Co-author** Responsible for TGA thermal measurement

Conferences & Training

Presenter **2026 TwIChE Annual Meeting**
2026/12/05–06 • Feng Chia University
Title: *Weakly Solvating Allyl Ether Electrolytes for Subzero Li ‖ SPAN Batteries: Solvation Structure, Transport Properties and Separator Wettability*

Vendor seminar **TA Instruments ARES-G3 launch & applications forum**
2026/07/17 • National Taiwan University

Vendor lecture **TA Instruments TGA lecture**
2026/06/03 • National Taiwan University

EHS Licenses

- Dedicated Wastewater and Sewage Treatment Specialist (Class B)
  2026 / 06 · certified · No. GB040231
- Indoor Air Quality Maintenance Management Specialist
  2026 / 08 · certified · No. IA020236
- Waste Disposal Technician (Class A)
  2026 / 10 · certified · No. HA160629
- Occupational Safety and Health Management Personnel (Level B)
  2026 / 11 · exam, pre-screen passed
- Occupational Safety Management Specialist (Level A)
  2027 / 03 · exam scheduled

Industrial Hygiene Engineer credit programme (Chinese Culture University)
Industrial Hygiene and Industrial Safety completed; OSH Management & Risk Assessment in progress.

Education

2025 – 2027/08 (expected)
National Taiwan University of Science and Technology
M.S. in Chemical Engineering
Low-temperature electrolytes and battery aging; coursework incl. Process Safety & Radiation Protection.

2020 – 2024
Tunghai University
B.S. in Chemical & Materials Engineering

Skills & Instruments

EHS fundamentals
GHS labelling · Ventilation · LEL / UEL · Exposure limits

Process & operations
Equipment procurement & maintenance

Software
Aspen Plus V14 · Aspen Dynamics · Microsoft Office · OriginPro · TA Universal Analysis · TA TRIOS

Instruments

Maintenance
TGA Q500 · DSC 2500

Operation
Anton Paar DMA 5002/Lovis 2001 · ARES-G3 · Glovebox

Procurement
Glovebox O_2 sensor · DSC sample press · Hydraulic electrode punch · Vacuum pump · Eppendorf Pipette · Low-temperature bath

NSRRC synchrotron
TPS 45A2 Submicron Soft X-ray Spectroscopy
TPS 27A2 Soft X-ray Nanoscopy
TLS 20A1 BM-(H-SGM) XAS
Assisted measurements; observed upkeep & evacuation routes